

BOUNDARY-VALUE PROBLEMS OF FUNCTIONAL DIFFERENTIAL EQUATIONS WITH STATE-DEPENDENT DELAYS*

ALESSIA ANDÒ[†] AND JAN SIEBER[‡]

Abstract. We prove convergence of piecewise polynomial collocation methods applied to periodic boundary value problems for functional differential equations with state-dependent delays. The state dependence of the delays leads to nonlinearities that are not locally Lipschitz continuous, preventing the direct application of general abstract discretization theoretic frameworks. We employ a weaker form of differentiability, which we call mild differentiability, to prove that a locally unique solution of the functional differential equation is approximated by the solution of the discretized problem with the expected order.

Key words. functional differential equations, delay differential equations, periodic solutions, boundary-value problems, collocation methods, state-dependent delay

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1. Introduction. Tracking time-periodic responses (*periodic orbits*) is a common task in numerical bifurcation analysis of nonlinear dynamical systems [34, 24]. The interest extends to periodic orbits that are dynamically unstable or extremely sensitive to parameters, as these orbits are thresholds between alternative stable states or are the link between seemingly discontinuous system responses, such as canards [14]. Dynamically unstable or sensitive orbits are beyond the reach of initial-value problem (IVP) solvers due to sensitivity to initial conditions near the periodic orbit. If the dynamical system is described by ordinary differential equations (ODEs), there are robust tools available that address sensitivity caused by dynamical instability and are able to track families of periodic orbits in one or many system parameters. Widely adopted tools for ODEs are AUTO [17], MATCONT [24], or COCO [13]. These tools compute solutions $y(t) \in \mathbb{R}^{n_y}$ of an ODE with parameters, $y'(t) = G(y(t), p)$ with an unknown period T , such that $y(t) = y(t + T)$ for all $t \in \mathbb{R}$ and some unknown $T > 0$. They solve the boundary-value problem (BVP) numerically with piecewise polynomial collocation similar to that described by Ascher, Christiansen, and Russell [5], where the time-rescaled periodic orbit $t \mapsto y(t/T)$ is approximated by a piecewise polynomial $s \mapsto y^L(s)$ for $s \in [0, 1]$ with L pieces of (usually uniform) degree m on L subintervals of $[0, 1]$. The method imposes the ODE at chosen time points (nodes)

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[†]CDLab – Computational Dynamics Laboratory, Department of Mathematics, Computer Science and Physics – University of Udine, 33100 Udine UD, Italy (alessia.ando@uniud.it).

[‡]Department of Mathematics and Statistics, University of Exeter, Exeter EX4 4QF, UK (j.sieber@exeter.ac.uk).

within each subinterval to construct a large nonlinear system of algebraic equations with a blockdiagonal Jacobian. Tools such as AUTO, MATCONT, and COCO *embed* the BVP by including one or several parameters into the unknown and augmenting the BVP with constraints (typically affine), such as phase and pseudo-arclength conditions. This augments the blockdiagonal Jacobian, resulting in well-conditioned problems that would be ill-conditioned if a shooting approach over a bounded number of time intervals was employed instead. See Desroches et al. [14] for several impressive demonstrations of how to find phenomena that occur in exponentially small parameter regions in singularly perturbed problems.

Functional differential equations. If the right-hand side of the differential equation depends on values of y at times other than the current t , one speaks of functional differential equations (FDEs), writing

$$(1.1) \quad \dot{y}(t) = G_{\text{FDE}}(y_t, p), \text{ where } G_{\text{FDE}} : C^0([-\tau_{\max}, 0]; \mathbb{R}^{n_y}) \times \mathbb{R}^{n_p} \rightarrow \mathbb{R}^{n_y},$$

where the dependent variable is $y : \mathbb{R} \rightarrow \mathbb{R}^{n_y}$, $p \in \mathbb{R}^{n_p}$ are the problem parameters, and G_{FDE} is a continuous nonlinear functional. The most common example class for equations such as (1.1) are delay differential equations (DDEs). Our illustrative example (1.3) below is a DDE. We use $C^\ell(I; \mathbb{R}^n)$ for the space of continuous (for $\ell = 0$) or continuously differentiable (for $\ell > 0$) functions from the interval $I \subset \mathbb{R}$ into $\mathbb{R}^n$. The subscript t in y_t denotes the time shift operator

$$C^0(\mathbb{R}; \mathbb{R}^{n_y}) \times \mathbb{R} \ni (y, t) \mapsto y_t \in C^0([-\tau_{\max}, 0]; \mathbb{R}^{n_y}) \text{ with } y_t(s) = y(t + s)$$

(we may also write $\dot{y}(t) = G_{\text{FDE}}(y(t + (\cdot)), p)$). Common challenging bifurcation analysis problems involving FDEs arise in optics due to transport delays [42], in population dynamics due to nonzero maturation times (leading to implicitly defined threshold delays and integrals over the past [21, 15]), or in machining due to the effects of the machining tool on the surface from the previous revolution [29].

Collocation for FDEs. As the piecewise polynomial y^L provides a natural interpolation, the collocation methods for ODEs immediately generalize to FDEs. After time rescaling, one is looking for a 1-periodic function satisfying the FDE

$$(1.2) \quad y'(t) = TG_{\text{FDE}}(y(t + (\cdot)/T), p),$$

such that one may evaluate the functional G_{FDE} at a collocation point t using the piecewise polynomial $(y^L)'(t) = TG_{\text{FDE}}(y^L(t + (\cdot)/T), p)$. Imposition of the FDE at a time point $t \in [0, 1]$ introduces coupling between values of y at different times such that the Jacobian of the resulting nonlinear system of algebraic equations is no longer blockdiagonal. When one seeks to find periodic orbits, one may “wrap around” when finding the value of y at deviating arguments $t + s$ outside the base interval $[0, 1]$ by using $(t + s) \bmod 1$. So the nondiagonal coupling is the only added difficulty when formulating discretized periodic BVPs for FDEs. This motivates specialized BVP solvers and analysis for the case of finding periodic orbits in FDE problems with parameters.

Complete tools for bifurcation analysis incorporating collocation for periodic orbits were developed and implemented by Engelborghs, Luzyanina, and Roose [20] (DDE-BIFTOOL) and Szalai [45] (KNUT); see also [41] for a review. These tools permit an arbitrary number of discrete delays as part of G_{FDE} , which may depend on the state and system parameters (for DDE-BIFTOOL). A posteriori convergence tests on examples suggested convergence orders equal to the degree m of the polynomial

pieces for the maximum norm of the error over interval $[0, 1]$. One cannot expect better (e.g., superconvergence), as the interpolation uses the piecewise polynomial $y^L(\cdot)$ of degree m when evaluating G_{FDE} at the collocation points [19, 6]. Engelborghs and Doedel [18] proved linear stability for collocation methods for time-periodic linear inhomogeneous FDEs with discrete constant delays. They pointed to “general stability theory for discretizations of nonlinear operator equations” for concluding (informally) convergence of the methods, referring to [30]. An alternative are methods based on series expansions of Chebyshev or Fourier type, thus fixing $L = 1$, letting m go to infinity, and choosing a projection of the right-hand side onto the truncated series up to m . The high-order a posteriori convergence observed for these methods makes them excellent candidates for verified numerical computations. Gimeno and co-workers demonstrated how Chebyshev series truncations can be applied to some DDEs with a single state-dependent delay to prove existence of periodic orbits and their isochrons with verified numerics [22, 23].

Lack of continuous differentiability of the nonlinearity. However, the argument by Engelborghs and Doedel [18] is only valid if one treats the period T of the orbit and the delays (which are system parameters) as known constants.

Let us illustrate the obstacle to applying standard arguments for convergence of numerical methods with the simple example

$$(1.3) \quad \dot{y}(t) = -y(t - p - y(t)),$$

where $p \approx \pi/2$. This FDE fits the general form (1.1) with $G_{\text{FDE}}(y, p) = -y(-p - y(0))$, which is continuous on an open subset of $C^0([-\tau_{\max}, 0]; \mathbb{R}) \times \mathbb{R}$ for $\tau_{\max} > \frac{\pi}{2}$ (so $n_y = n_p = 1$). The Hopf bifurcation theorem ensures that this FDE has a family of small-amplitude periodic solutions of the form $y(t) = y_0 \sin(\theta + t/T) + O(|y_0|^2)$ with $0 < |y_0| \ll 1$ and period $T = 2\pi + O(|y_0|^2)$ for $p = \pi/2 + O(|y_0|^2)$ and arbitrary θ . See [34] for the classical Hopf bifurcation theorem, [25, 16] for the version for FDEs with constant delay, and [43] for its proof for FDEs with state-dependent delays. After rescaling the time interval to $[0, 1]$, the unknown period T appears explicitly: as a parameter, we look for 1-periodic functions y , periods T , and parameters p such that

$$(1.4) \quad y'(t) = G(y_t, T, p) = -Ty(t - p/T - y(t)/T) \text{ for all } t \in [0, 1].$$

For the rescaled problem (1.4), the right-hand-side nonlinearity has the form

$$(1.5) \quad G_{\text{ts}} : C_\pi^\ell \times \mathbb{R} \times \mathbb{R} \ni (y(\cdot), T, p) \mapsto Ty((\cdot) - p/T - y(\cdot)/T) \in C_\pi^\ell,$$

where the subscript $(\cdot)_{\text{ts}}$ stands for *time shift* and we use C_π^ℓ for spaces of ℓ times continuously differentiable 1-periodic functions. We observe that unknowns appear inside the arguments of y , which is itself unknown, such that application of the derivative to G_{ts} reduces the regularity of the argument y of G_{ts} :

$$(1.6) \quad \begin{aligned} DG_{\text{ts}}(y, T, p)[\delta^y, \delta^T, \delta^p](t) \\ = -T\delta^y(t - p/T - y(t)/T) + y'(t - p/T - y(t)/T)[\delta^y(t) + \delta^p]/T \\ - \delta^T[y(t - p/T - y(t)/T) + y'(t - p/T - y(t)/T)[p + y(t)]/T \end{aligned}$$

for a deviation $(\delta^y, \delta^T, \delta^p) \in C_\pi^\ell \times \mathbb{R} \times \mathbb{R}$. Thus, the right-hand side G_{ts} is not differentiable or locally Lipschitz continuous if we consider G_{ts} as mapping $y \in C_\pi^\ell$ into C_π^ℓ for any $\ell \geq 0$. The review in [27] pointed out this lack of continuous differentiability and its consequences (see also [12]). For example, solutions for IVPs are not unique

if one permits C^0 initial conditions. If the initial conditions are at least C^1 and *compatible* [48], then the dependence of IVP solutions on initial values is C^1 , but results on higher regularity are missing. Large parts of the theory for FDEs as developed in textbooks by Hale [25], Hale and Verduyn Lunel [26], and Diekmann et al. [16] rely on continuous differentiability of G_{ts} acting on arguments $y \in C^0$ and are thus not applicable to problems such as (1.3).

Convergence of numerical discretization methods for BVPs. Similarly, discretization theory for boundary-value problems (BVPs) has been developed for nonlinearities G_{ts} that are differentiable for C^0 arguments by Maset [36, 37, 38]. Indeed, it is tempting to assume differentiability of the nonlinearities and their discretized counterparts in FDE BVPs because theorems on zeros of differentiable operators such as [32, Lemma 19.1] allow one to prove the uniqueness of the fixed point of the discretized problem straightforwardly. Andò and Breda [3] pointed out that for periodic BVPs with unknown period T the rescaling by the unknown period T introduces a state dependence of the deviating time arguments (namely on T ; see the term $Ty(t - p/T \dots)$ in (1.4)), even for FDE problems with constant delay. This causes a loss of differentiability of the problem's nonlinearity with respect to the unknown T . Careful reanalysis of the general framework constructed by Maset [38] showed that continuous differentiability with respect to the scalar variable T is only needed in a single point, namely the assumed-to-exist solution of the BVP. Thus, Andò and Breda [3] proved convergence of the routinely used methods in DDE-BIFTOOL and KNUT for FDEs with constant delays for the first time, closing the gap left in the argument of [18].

The analysis of Andò and Breda [3] leaves the question open as to how necessary special treatment of the unknown period T is (a finite-dimensional part of the unknowns of the problem) or if convergence of numerical discretization for periodic BVPs can be proved without assuming continuous differentiability of the right-hand side (also referred to as *Fréchet differentiability* in, e.g., [1]). The review in [27] points to the appropriate generalized differentiability properties that are satisfied by the nonlinearities occurring in FDEs. As one can see in the right-hand side (1.5) and its derivative (1.6), nonlinearities in FDEs are differentiable ℓ times with respect to their arguments y and p only if $y \in C^\ell$. In (1.6), we also see that while the derivative depends on y' , it does not depend on $(\delta y)'$. We use this property of restricted differentiability (we call it *mild* in Definition 2.1) to prove our central convergence result for discretizations of embedded periodic BVPs of FDEs. In particular, our main result is the error estimate $O(L^{-\min\{\ell_{\max}, m\}})$ for right-hand sides which are mildly differentiable to order $\ell_{\max}$, where m is the fixed degree of the approximating piecewise polynomials and L is the increasing size of the mesh.

2. Main results.

2.1. Periodic BVPs for FDEs and mild differentiability. Consider an embedded periodic BVP of the form

$$(2.1) \quad y'(t) = TG_{\text{FDE}}(y(t + (\cdot)/T), p), \quad 0 = R_{\text{aff}}[y, T, p]$$

for continuous 1-periodic $y : \mathbb{R} \rightarrow \mathbb{R}^{n_y}$, period T , and parameter $p \in \mathbb{R}^{n_p}$, where $G_{\text{FDE}} : C^0([-\tau_{\max}, 0]; \mathbb{R}^{n_y}) \times \mathbb{R}^{n_p} \rightarrow \mathbb{R}^{n_y}$ is continuous. The affine map $R_{\text{aff}} : C_\pi^0 \times \mathbb{R}^{n_p+1} \rightarrow \mathbb{R}^{n_p+1}$ defines constraints, making the system “square” when including the parameters T and p as unknowns. The constraints should also eliminate the shift symmetry that $y(\theta + (\cdot))$ is a solution of (1.3) for all θ whenever it is a solution for

$\theta = 0$. Nonlinearities such as G_{FDE} require the concept of *mild differentiability* to describe their regularity.

DEFINITION 2.1 (mild differentiability). *A functional $G : C^0 \rightarrow \mathbb{R}^{n_G}$ is called $\ell_{\max}$ times mildly differentiable if the following hold:*

1. (restricted continuous differentiability) *its restriction $G|_{C^\ell} : C^\ell \rightarrow \mathbb{R}^{n_G}$ is ℓ times differentiable for $\ell \leq \ell_{\max}$ (in particular, G is continuous); and*
2. (extendability) *the map $C^\ell \times C^\ell \ni (u, \delta^u) \mapsto D^\ell G(u)(\delta^u)^\ell \in \mathbb{R}^{n_G}$ has a continuous extension to $C^\ell \times C^{\ell-1}$ for all $\ell \leq \ell_{\max}$, where $(\delta^u)^\ell$ is the ℓ -tuple $\delta^u \dots \delta^u$ of arguments of the multilinear map $D^\ell G(u)$.*

This definition extends naturally to the functional on the right-hand side of (2.1),

$$G(y, T, p) := TG_{\text{FDE}}(y(t + (\cdot)/T), p),$$

through the embedding of $\mathbb{R}^{n_p+1}$ into C_π^0 that treats a vector (T, p) as the constant function $t \mapsto (T, p)$. We list basic properties of mildly differentiable functionals in section 4, which will be needed for our convergence analysis in section 6. These include the validity of the chain rule, which ensures that $t \mapsto TG_{\text{FDE}}(y(t + (\cdot)/T), p)$ is in C^ℓ if $y \in C^\ell$ and G_{FDE} is ℓ times mildly differentiable. The nonlinearity $(y, T, p) \mapsto -Ty(-p/T - y(0)/T)$ used in example (1.4) is an example of a mildly differentiable functional (to arbitrary degree). Definition 2.1 extends the classical “mild regularity assumption” made to obtain classical results for state-dependent FDEs [48, 27] to arbitrary degree $\ell > 1$. Alternative approaches taken in [33, 43] are equivalent but more involved to state and check in practical examples. All problems that can be formulated with standard numerical tools such as DDE-BIFTOOL and DDESD available in MATLAB satisfy mild differentiability in the sense of Definition 2.1 if the problem coefficient functions are sufficiently regular. See [28] for a demonstration of how DDE-BIFTOOL can be used for FDEs with state-dependent delays and mildly differentiable right-hand sides.

Assumption 2.2 (assumptions on the BVP).

1. (Mild differentiability) The right-hand side G_{FDE} in the FDE (2.1) is mildly differentiable to order $\ell_{\max} \geq 1$.
2. (Existence of solution) BVP (2.1) has a solution (y^*, T^*, p^*) .
3. (Linear well-posedness) The BVP (2.1), linearized in (y^*, T^*, p^*) , has only the trivial solution $(\delta^y, \delta^T, \delta^p) = 0$.

Corollary 5.3 below shows that points 1 and 2 imply that $y^* \in C_\pi^2$ such that the derivative $DG(y^*, T^*, p^*)$ exists for the mildly differentiable G , which permits us to pose Assumption 2.2, point 3. Section 5 will restate Assumption 2.2 for a fixed-point problem equivalent to (2.1), defined later in (3.4).

2.2. Convergence of solution of discretized BVP. For polynomial collocation, the unknown function is a 1-periodic continuous piecewise polynomial y^L on a mesh given as $0 = t_0 < \dots < t_L = 1$. More precisely, y^L is a polynomial of degree m on $[t_{i-1}, t_i]$ for all $i = 1, \dots, L$ in each of its n_y components, y^L is continuous, and $y(t) = y(t+1)$ for all t . Additional unknowns are the parameters (T^L, p^L) , resulting in $n_y mL + n_p + 1$ unknowns overall. The system of algebraic equations

$$(2.2) \quad 0 = (y^L)'(t_{i,j}) - T^L G_{\text{FDE}}(y^L(t_{i,j} + (\cdot)/T^L), p^L), \quad 0 = R_{\text{aff}}[y^L, T^L, p^L]$$

for $1 \leq i \leq L$, $1 \leq j \leq m$, evaluates the FDE at the collocation points $t_{i,j} = t_{i-1} + (t_i - t_{i-1})t_{c,j}$, where the points $t_{c,j}$ are the m collocation points for degree $m-1$ on

the interval $[0, 1]$ for a sequence of orthogonal polynomials (e.g. of Gauss–Legendre or Chebyshev type). The strategy for adjusting approximation quality is a finite-element approach [2, 3], keeping the degree m bounded and considering the limit $L \rightarrow \infty$, refining the mesh such that $\max\{t_i - t_{i-1}\} \leq C_{\text{msh}}/L$ for a bounded C_{msh} independent of L . In contrast, the strategy of keeping L bounded and letting m go to infinity is called the *spectral* element method in [9, 46]. Spectral methods promise exponential convergence but require bounds on all derivatives of y^* . In the proofs of Lemmas 6.1 and 6.5, we rely on the boundedness of the degree m , but both lemmas only require G to be mildly differentiable once.

We can now state a convergence theorem for the discretized BVP (2.2).

THEOREM 2.3 (convergence of discretization). *Under Assumption 2.2, the discretized BVP (2.2) with $\max\{t_i - t_{i-1}\} \leq C_{\text{msh}}/L$ has a locally unique solution $x_L = (y^L, T^L, p^L)$ near $x^* = (y^*, T^*, p^*)$ for all sufficiently large L , satisfying*

$$(2.3) \quad \|x_L - x^*\|_{0,1} := \max\{\|y^L - y^*\|_{0,1}, |T^L - T^*|, |p^L - p^*|\} = O(L^{-\min\{\ell_{\max}, m\}}).$$

The norm $\|y^L - y^*\|_{0,1}$ in (2.3) is the Lipschitz norm of the discretization error.

Proof through equivalent fixed-point problem. Andò and Breda (see [2, 3]) reformulate the BVP (2.1) as a fixed-point problem, following the general framework in [38]. We modify this approach in section 3 to exploit the special structure present in periodic BVPs by constructing a fixed-point problem where the right-hand side is compact and maps spaces of periodic functions into spaces of periodic functions.

For this fixed-point problem, the discretization corresponds to inserting a projection operator $\mathcal{P}_L$ onto the space of discontinuous piecewise polynomials of degree $m - 1$ in between the nonlinearity and an integral operator. This formulation ensures that the space of numerical solutions y^L consists of continuous 1-periodic piecewise polynomials of degree m . Theorem 6.6 in section 6 establishes convergence for the discretized fixed-point problem.

Mild differentiability of G_{FDE} helps us in section 6 to establish the consistency and stability of the numerical method and the smallness of the nonlinear terms, leading to the proof of Theorem 6.6 and hence Theorem 2.3.

2.3. Illustrative example of BVP. We will use the example (1.3) throughout to illustrate concepts and results. The embedded BVP for finding periodic orbits of (1.3), rescaled to base interval $[0, 1]$ with additional affine conditions, is

$$(2.4) \quad y'(t) = -Ty(t - (p + y(t))/T), \quad (y(t) = y(t + 1) \text{ for all } t \in \mathbb{R}),$$

$$(2.5) \quad 0 = R_{\text{aff}}[y, T, p] := \begin{bmatrix} y(0) \\ 2 \int_0^1 \sin(2\pi t) y(t) dt - y_0 \end{bmatrix}$$

for a range of small y_0 . It is sufficient to impose (2.4) on the base interval $[0, 1]$ if one takes into account the periodicity of y . For the choice (2.5) of R_{aff} , the first condition fixes the phase of the solution y (serving as a phase condition) and the second condition fixes the amplitude, locally parametrizing the solution family by the constant y_0 (thus serving as a pseudo-arclength condition). System (2.4)–(2.5) will have solutions of the form $y(t) = y_0 \sin(2\pi t) + O(|y_0|^2)$, $T = 2\pi + O(|y_0|^2)$, $p = \pi/2 + O(|y_0|^2)$ for small y_0 according to the Hopf bifurcation theorem [43]. The unknowns in this problem are $(y(\cdot), T, p)$.

The functional $G(y, T, p)$ for this example and its derivative (defined on $C_\pi^1 \times \mathbb{R}^2$) have already been used for illustration in (1.4) and in (1.6).

3. Fixed-point problem equivalent to periodic BVP and relevant notation. In the formulation (2.1), the unknown period T plays the same role as a problem parameter. Thus, in the following we collect T and p into an $(n_p + 1)$ -dimensional parameter vector $\mu = (T, p)$, introducing the nonlinear functional

$$(3.1) \quad G(y, \mu) = TG_{\text{FDE}}(y((\cdot)/T), p), \quad \text{where now } \mu = (T, p) \in \mathbb{R}^{n_\mu}, \quad n_\mu = n_p + 1,$$

and abbreviate $R_{\text{aff}}[y, \mu] = R_{\text{aff}}[y, T, p]$ to simplify notation. Thus, (2.1) is a special case of the general BVP

$$(3.2) \quad \begin{aligned} y'(t) &= G(y_t, \mu) && \in \mathbb{R}^{n_y} \text{ for } t \in [0, 1) && \text{(FDE for 1-periodic continuous } y), \\ 0 &= R_{\text{aff}}[y, \mu] && \in \mathbb{R}^{n_\mu} && \text{(affine constraints).} \end{aligned}$$

We reformulate (3.2) as a fixed-point problem for the solution, obtained from the differential equation through the variation-of-constants formula, following [3],

$$(3.3) \quad y(t) = y(0) + \int_0^t G(y_s, \mu) \, ds.$$

We plan to pose the fixed-point problem in a space of periodic functions of period 1. However, the right-hand side of (3.3) is not guaranteed to be 1-periodic even if y is 1-periodic. To enforce period 1, we subtract the average of the integrand and then impose that this average is zero as a separate equation, replacing the periodic boundary condition. We also introduce the new variable $\alpha \in \mathbb{R}^{n_y}$, which will be equal to $y(0)$ in the solution. Hence, BVP (2.1) is equivalent to the fixed-point problem

$$(3.4) \quad x = \Phi(x)$$

for the operator

$$(3.5) \quad \Phi(x) := \begin{bmatrix} t \mapsto \alpha + \int_0^t G(v_s, \mu) \, ds - t \int_0^1 G(v_s, \mu) \, ds \\ \alpha + \int_0^1 G(v_s, \mu) \, ds \\ \mu + R_{\text{aff}}[v, \mu] \end{bmatrix} \quad \text{for } x = \begin{bmatrix} v(\cdot) \\ \alpha \\ \mu \end{bmatrix},$$

where v is 1-periodic with $v(t) \in \mathbb{R}^{n_y}$, $\alpha \in \mathbb{R}^{n_y}$, $\mu \in \mathbb{R}^{n_\mu}$. Fixed-point problem (3.4) is then equivalent to the original problem of finding a periodic solution of FDE (1.1) in the sense that for every periodic solution $y(t)$ with period $T > 0$ of (1.1) at parameter p , there exists a phase shift $\theta \in \mathbb{R}$ such that $x = (t \mapsto y(\theta + tT), y(\theta), T, p)$ is a solution of (3.4), and vice versa; for a fixed point $x = (v, \alpha, \mu) = (v, \alpha, T, p)$ of Φ , the function $y(t) = v(t/T)$ is a periodic solution of (1.1) with period T at parameter p , α must be equal to $v(0)$, and (v, T, p) satisfy the constraints $R_{\text{aff}}[v, T, p]$.

Notation for function spaces and norms. The definition of Φ in (3.5) had not specified the function space for the v component of x yet, as we have not discussed what type of perturbation a discretization may introduce for Φ . To specify suitable spaces, we use the notation

$$\begin{aligned} C_\pi^k &= \{v : k \times \text{cont. diff., } v(t) = v(t+1) \text{ for all } t\}, & \|v\|_k &= \max_{t \in [0,1], j \leq k} |v^{(j)}(t)|, \\ C_\pi^{k,1} &= \{v \in C_\pi^k, \text{Lip } v^{(k)} < \infty\}, & \|v\|_{k,1} &= \max\{\|v\|_k, \text{Lip } v^{(k)}\}, \\ L_\pi^\infty &= \{v \text{ ess.bd., } v(t) = v(t+1) \text{ for all } t\}, & \|v\|_\infty &= \text{ess sup}_{t \in [0,1]} |v(t)|. \end{aligned}$$

The dimension of the function's value $v(t)$ is determined by context such that we often drop domain and codomain indicators in the spaces. Otherwise, we write, e.g., $C_{\pi}^{k,j}(\mathbb{R}^{n_u})$ or $L_{\pi}^{\infty}(\mathbb{R}^{n_u})$.

Linear and nonlinear part of fixed-point problem $x = \Phi(x)$. The variable x has several components, $x = (v, \alpha, \mu)$, introduced above, of which only the first one, v , is infinite-dimensional such that all norms of spaces for v can be trivially extended by the finite-dimensional maximum norms of α and μ . Hence, we define the extended spaces

$$C_e^{k,j} = C_{\pi}^{k,j} \times \mathbb{R}^{n_y} \times \mathbb{R}^{n_{\mu}}, \quad L_e^{\infty} = L_{\pi}^{\infty} \times \mathbb{R}^{n_y} \times \mathbb{R}^{n_{\mu}}$$

and continue to use the $\|\cdot\|_{k,j}$ or $\|\cdot\|_{\infty}$ notation for their respective norms. We split the operator Φ , defined in (3.5), into a compact linear part $\mathcal{L}$ and a nonlinear part g , such that $\Phi = \mathcal{L} \circ g$. For $\mathcal{L}$ and g , we can now pick specific spaces:

$$(3.6) \quad g: C_e^{0,1} \rightarrow C_e^0, \quad \text{with} \quad g \left(\begin{bmatrix} v \\ \alpha \\ \mu \end{bmatrix} \right) = \begin{bmatrix} t \mapsto G(v_t, \mu) \\ \alpha \\ \mu + R_{\text{aff}}[v, \mu] \end{bmatrix}, \quad \text{and}$$

$$(3.7) \quad \mathcal{L}: L_e^{\infty} \rightarrow C_e^{0,1}, \quad \text{with} \quad \mathcal{L} \begin{bmatrix} w \\ \alpha \\ \nu \end{bmatrix} = \begin{bmatrix} t \mapsto \alpha + \int_0^t w(s) \, ds - t \int_0^1 w(s) \, ds \\ \alpha + \int_0^1 w(s) \, ds \\ \nu \end{bmatrix}.$$

The convergence theorem (Theorem 6.6) for fixed-point problem (3.4) proves that the fixed point x^L of a discretized operator Φ_L converges with the rate expected by the order of the discretization scheme to a fixed point x^* of Φ under appropriate conditions. This convergence will operate on a small ball $B_r^{0,1}(x^*)$ of Lipschitz continuous functions in $C_e^{0,1}$ around x^* . The center x^* has higher-order regularity than $C_e^{0,1}$: $x^* \in C_e^{\ell_{\max}+1}$, where $\ell_{\max} \geq 0$ depends on regularity assumptions on the right-hand side G .

Discretized fixed-point problem. For functions $z \in C_{\pi}^0$, we define the interpolation projection $P_L z$ as the unique piecewise polynomial on mesh $(t_i)_{i=0}^L$ of degree $m-1$ where the piece on each interval (t_{i-1}, t_i) equals z on the nodes $t_{i,j}$:

$$\begin{aligned} P_L: C_{\pi}^0 &\mapsto L_{\pi}^{\infty}, \\ P_L z &= \hat{z} \quad \text{with } \hat{z}(t_{i,j}) = z(t_{i,j}) \text{ for all } i \in \{1, \dots, L\}, j \in \{1, \dots, m\}, \\ &\quad \text{and } \hat{z} \text{ is degree } m-1 \text{ polynomial on } (t_{i-1}, t_i) \text{ for all } i \in \{1, \dots, L\}. \end{aligned}$$

In the name P_L , we do not indicate the dependence on the interpolation degree $m-1$, as we will keep this degree constant, studying only the limit $L \rightarrow \infty$ in our convergence analysis. The dependence on the mesh (t_i) , which will change with increasing L , is also implicitly included in the subscript L . The interpolating piecewise polynomial $t \mapsto [P_L z](t)$ is not necessarily continuous, as it may have discontinuities at the mesh boundaries t_i , such that the codomain of P_L is L_{π}^{∞} . By the construction of $\Phi = \mathcal{L} \circ g$ and P_L , we have the following equivalence.

LEMMA 3.1 (equivalence of discretized fixed-point problem). *The discretized BVP (2.2) is equivalent to the fixed-point problem*

$$(3.8) \quad x = \mathcal{L} P_L g(x) =: \Phi_L(x),$$

where $\mathcal{L} : L_e^\infty \rightarrow C_e^{0,1}$ and $g : C_e^{0,1} \rightarrow C_e^0$ are defined in (3.6) and (3.7), and $\mathcal{P}_L$ is the trivial extension of P_L ,

$$(3.9) \quad \mathcal{P}_L : C_e^0 \rightarrow L_e^\infty, \quad \mathcal{P}_L \begin{bmatrix} w \\ \alpha \\ \nu \end{bmatrix} = \begin{bmatrix} P_L w \\ \alpha \\ \nu \end{bmatrix}.$$

Remark 3.2 (discretized solution space). In our notation, the discretized fixed-point operator $\Phi_L = \mathcal{L} \circ \mathcal{P}_L \circ g$ has the codomain

$$\text{rg } \mathcal{LP}_L = \{p \in C_\pi^{0,1}, p|_{[t_{i-1}, t_i]} \text{ } m\text{-degree poly. for } i = 1, \dots, L\} \times \mathbb{R}^{n_y} \times \mathbb{R}^{n_\mu} \subset C_e^{0,1}.$$

In particular, if $x = (v, \alpha, \mu) \in \text{rg } \mathcal{LP}_L$, then its first component $v \in C_\pi^{0,1}$ is Lipschitz continuous but cannot be expected to be continuously differentiable.

Example. We illustrate Remark 3.2 for our rescaled example (1.4). The first component of $Dg(x)\delta^x$ in $x = (y, T, p)$ and $\delta^x = (\delta^y, \delta^T, \delta^p)$ is a function of time that has the form $DG_{\text{ts}}(y, T, p)[\delta^y, \delta^T, \delta^p]$ and is given in (1.6). It contains the term $y'(t - p/T - y(t)/T)$, which is not defined for all t if $y \in C_\pi^{0,1}$. In particular, when attempting to evaluate $DG_{\text{ts}}(y, T, p)$ for a function y in the discrete solution space $\text{rg } \mathcal{LP}_L$, one encounters an ill-defined term whenever a mesh point t_k and a collocation point $t_{i,j}$ satisfy

$$t_{i,j} - p/T - y(t_{i,j})/T - t_k \in \mathbb{Z} \quad \text{for some } i, k \in \{1, \dots, L\}, j \in \{1, \dots, m\}$$

because in the mesh boundary points t_k , the right-sided derivative and the left-sided derivative are generally different (see (2.2) for an introduction of collocation points $t_{i,j}$ and mesh points t_k).

4. Mildly differentiable nonlinear functionals.

Consequences of mild differentiability for nonlinearities of FDEs. Let $G : C_\pi^0 \mapsto \mathbb{R}^{n_G}$ be mildly differentiable to order $\ell_{\max} \geq 1$. We define $G_{\text{ts}} : C_\pi^0 \rightarrow C_\pi^0$ as

$$(4.1) \quad G_{\text{ts}}(u)(t) = G(u_t)$$

(domain and codomain have possibly different dimensions n_u and n_G), which combines the nonlinear functional $u \mapsto G(u)$ with the time shift $u \mapsto u_t$. The map g in our fixed-point problem (3.5) is of this type in its first component, $(v, \alpha, \mu) \mapsto [t \mapsto G(v_t, \mu)]$. The continuity of $G_{\text{ts}}(u)$ in time and the continuity of G_{ts} in u in the C^0 -norm follow from the continuity of $(t, u) \mapsto u_t$ and $G : C_\pi^0 \rightarrow \mathbb{R}^{n_G}$. The following lemmas collect relevant conclusions for this type of nonlinearity, which will be needed to obtain results on the regularity of a fixed point x^* of Φ as well as the convergence of the fixed point x^L of the discrete version Φ_L to x^* .

LEMMA 4.1 (extended differentiability of nonlinearity with time shift). *Assume that $G : C_\pi^0 \rightarrow \mathbb{R}^{n_G}$ satisfies mild differentiability to at least order $\ell_{\max} \geq 1$ according to Definition 2.1, and define $G_{\text{ts}}(u)(t) = G(u_t)$ for $u \in C_\pi^0$. Then for $\ell \leq \ell_{\max}$ the map $G_{\text{ts}} : C_\pi^{\ell_{\max}} \rightarrow C_\pi^{\ell_{\max} - \ell}$ is ℓ times continuously differentiable (for $\ell = 0$ continuous). The map*

$$C_\pi^{\ell_{\max}} \times (C_\pi^{\ell_{\max} - 1})^\ell \ni (u, \delta^{u,1}, \dots, \delta^{u,\ell}) \mapsto D^\ell G_{\text{ts}}(u) \delta^{u,1} \dots \delta^{u,\ell} \in C_\pi^{\ell_{\max} - \ell}$$

is continuous.

See Appendix A for a detailed proof for this lemma. The case $\ell_{\max} = 1$ for Lemma 4.1 illustrates where the extendability condition for G in the definition of mild differentiability is needed for differentiability of $G_{\text{ts}}(u)$ in time. Formally, one “applies the chain rule,” differentiating $G(u_t)$ with respect to t with $u \in C_\pi^1$ (for $k = 1$), such that $(G_{\text{ts}}(u))' = [G(u_t)]' = DG(u_t)u_t'$, but since $(u_t)' \in C_\pi^0$, the expression $DG(u_t)u_t'$ is only valid and continuous because of the continuous extendability of $DG(u_t)$ to C_π^0 (in which $(u_t)'$ lies). This is point 2 in Definition 2.1. The general case ($\ell_{\max} > 1$) then applies this argument repeatedly. In particular, we see that G_{ts} is only continuously differentiable ℓ times from a higher-regularity space $C_\pi^{\ell_{\max}}$ to the lower-regularity space $C_\pi^{\ell_{\max}-\ell}$.

We observe that the finite difference limit can be extended to Lipschitz continuous deviations (formulated and needed only for $\ell = 1$) by continuity.

COROLLARY 4.2 (extension of finite-difference limit). *Assume that $G : C_\pi^0 \rightarrow \mathbb{R}^{n_G}$ is mildly differentiable once. Then the map $C_\pi^1 \times C_\pi^0 \ni (u, \delta^u) \mapsto DG_{\text{ts}}(u)\delta^u \in C_\pi^0$ defined as in (4.1) satisfies for all $u \in C_\pi^1$*

$$(4.2) \quad \lim_{\substack{v, w \in C_\pi^{0,1} \\ \|v\|_{0,1}, \|w\|_{0,1} \rightarrow 0}} \frac{\|G_{\text{ts}}(u+v) - G_{\text{ts}}(u+w) - DG_{\text{ts}}(u)[v-w]\|_0}{\|v-w\|_{0,1}} = 0.$$

Note that in the finite-difference quotient limit (4.2), we extend from the C^1 -norm to the $C^{0,1}$ -norm. This is possible because the derivative DG of G (and hence the derivative DG_{ts} of G_{ts}) is continuously extendable to C_π^0 in its linear argument.

The regularity properties outlined in this section imply, in particular, that we cannot rely on the general framework by Maset [38] to prove the convergence of the numerical method, in contrast to [3], which studied the constant delay case. In other words, it is not possible to prove all the (theoretical and numerical) assumptions made in [38] to reach our convergence result. In particular, some assumptions will only hold in a weaker form, with different norms in the relevant inequalities than those in the general framework in [38]. In section 6, we will show that this does not impede high-order convergence for polynomial collocation methods, such that we obtain convergence results identical to those for constant delays.

5. Assumptions on the (infinite-dimensional) problem and immediate consequences. This section reformulates Assumption 2.2, stated in section 2 for the BVP, as assumptions on the infinite-dimensional fixed-point problem (3.5) for Φ . These are fewer than stated for the general theory by Maset [38] because Maset includes further assumptions needed for determining the convergence rate of the Newton iterations when solving the discretized problem.

Existence and regularity of solution to infinite-dimensional problem.

Assumption 5.1 (existence of solution). The fixed-point problem (3.5) has a solution x^* : $x^* = \Phi(x^*)$.

We denote the components of x^* as (v^*, α^*, μ^*) . By the construction of the fixed-point problem, v^* will be Lipschitz continuous and periodic with period 1 and will satisfy the differential equation

$$(5.1) \quad (v^*)'(t) = G(v_t^*, \mu^*),$$

such that v^* will even be in C_π^1 . The Hopf bifurcation theorem, proved for functional differential equations with state-dependent equations in [43], provides a scenario that

ensures the existence of periodic orbits and hence solutions of fixed-point problem (3.5). The illustrative example (1.3) is chosen to satisfy the assumptions of the Hopf bifurcation theorem for $p \approx \pi/2$ and small-amplitude periodic solutions $y(t)$ with period 2π (hence solutions $(y(\cdot), T, p)$ of (2.4), (2.5) with small-amplitude y).

Assumption 5.2 (mild differentiability of right-hand side G). The right-hand side G of the FDE (3.2) is mildly differentiable to order $\ell_{\max} \geq 1$.

As we include the parameters (treating constants as special cases of 1-periodic functions), the dimensions are $n_u = n_y + n_\mu = n_y + n_p + 1$ for the argument of G and $n_G = n_y$ for the value of G .

Lemma 4.1 implies the following corollary about the regularity of the solution x^* of the fixed-point problem (3.5), $x = \Phi(x)$.

COROLLARY 5.3 (regularity of solution of fixed-point problem (3.5)). *Let $x^* = (v^*, \alpha^*, T^*)$ be a solution of the fixed-point problem (3.5), and let the right-hand side G of the FDE (3.2) satisfy mild differentiability to order $\ell_{\max}$. Then the solution component v^* is in $C_{\pi}^{\ell_{\max}+1}$ as a function of time, and hence $x^* \in C_e^{\ell_{\max}+1}$.*

The statement of Corollary 5.3 follows from Lemma 4.1, applied to the case $\ell = 0$, and using that v^* satisfies the differential equation (5.1): for each j from 0 to $\ell_{\max}$, we have that $G(v_t^*, \mu^*)$ in C_{π}^j because v^* is in C_{π}^j by Lemma 4.1. Then, by the differential equation (5.1), $(v^*)'(t)$ is in C_{π}^j , such that v^* is in C_{π}^{j+1} .

Hence, by Assumption 5.2 that $\ell_{\max} \geq 1$, we have that $x^* \in C_e^{\ell_{\max}+1} \subseteq C_e^2$. Note that Assumption 5.2 of mild differentiability of G with $\ell_{\max} = 1$, together with Corollary 5.3, implies that the derivative

$$C_{\pi}^1 \ni x \mapsto D\Phi(x) = \mathcal{L}Dg(x) \in \text{Lin}(C_{\pi}^1; C_{\pi}^1)$$

of the fixed-point map Φ defined in (3.5) exists and is continuous in $x = x^*$. This follows from the continuous differentiability of g as a map from C_e^1 to C_e^0 and the subsequent application of the linear continuous mapping $\mathcal{L} \in \text{Lin}(C_e^0; C_{\pi}^1)$, which increases regularity by one degree.

The last assumption that we make on the infinite-dimensional problem is the well-posedness of the system linearized around the fixed point, which will be needed to show the stability of the discretized problem.

Assumption 5.4 (well-posedness of infinite-dimensional linear problem). The linear bounded operator $I - D\Phi(x^*)$ is injective on C_e^1 ; that is, if $y = D\Phi(x^*)y$ and $y \in C_e^1$, then $y = 0$.

We denote the norm of the inverse

$$(5.2) \quad C_{\text{stab}, \infty} = \|[I - D\Phi(x^*)]^{-1}\|_{C_e^1 \leftarrow C_e^1} = \|[I - D\Phi(x^*)]^{-1}\|_{C_e^{0,1} \leftarrow C_e^{0,1}}.$$

Elements y of the nullspace of $[I - D\Phi(x^*)]$ that are at least in C_e^0 satisfy the identity $y = \mathcal{L}Dg(x^*)y$, such that they are in the image of $\mathcal{L}$ of C_e^0 , which is in C_{π}^1 , since $\mathcal{L}$ involves an antiderivative. Hence, we may replace C_e^1 in Assumption 5.4 by C_e^0 (and even L_e^{∞}).

Since $D\Phi(x^*)$ is a compact linear operator on spaces $C_e^{k,j}$ for all $k \geq 0$ and $j = 0, 1$, the nullspace of $D\Phi(x^*)$ is at most finite-dimensional, and $\dim \ker[I - D\Phi(x^*)] = 0$ implies the existence of a bounded inverse $[I - D\Phi(x^*)]^{-1}$. A sequence of results formulate Assumption 5.4 in terms of requiring full rank of a finite-dimensional *characteristic* matrix. One example construction for a characteristic matrix of a linear

time-periodic FDE is given in [44], generalized to FDEs of neutral type by Verduyn Lunel [47]. In practice, numerical methods for determining eigenvalues of monodromy operators, such as the pseudospectral method [8, 10, 11], work directly on large matrices generated by the discretization projection $\mathcal{P}_L$, without prior reduction. Borgioli et al. [7] prove the convergence of methods as implemented in DDE-BIFTOOL and KNUT.

6. Convergence of solutions of discretized problem. In this section, we show that solutions x of the fixed-point problem for the discretized map given in the equivalence lemma (Lemma 3.1),

$$(6.1) \quad \Phi_L = \mathcal{L}\mathcal{P}_L g,$$

are locally unique, and converge to the fixed point x^* of Φ (solving $x^* = \Phi(x^*) = \mathcal{L}g(x^*)$). In x^* (which is in $C_e^2 \subset C_e^1$), the derivative of Φ_L is well defined and equal to $D\Phi_L(x^*) = \mathcal{L}\mathcal{P}_L Dg(x^*)$. The discretized fixed-point problem $\Phi_L(x) = x$ can be formulated in terms of $\delta^x := x - x^*$ as

$$(6.2) \quad [I - D\Phi_L(x^*)]\delta^x$$

$$(6.3) \quad = \mathcal{L}(\mathcal{P}_L - I)g(x^*) \quad (\text{consistency term } \epsilon_c(L))$$

$$(6.4) \quad + \mathcal{L}\mathcal{P}_L[g(x^* + \delta^x) - g(x^*) - Dg(x^*)\delta^x] \quad (\text{nonlinearity term } \epsilon_{nl}(L, \delta^x)).$$

Given the left-hand side of (6.2), a necessary condition for the sought well-posedness is the invertibility of the operator $I - D\Phi_L(x^*)$ uniformly for $L \rightarrow \infty$, i.e., the stability of the discretized problem. This is particularly evident in the case of a linear Φ , where $\epsilon_{nl}(L, \delta^x) = 0$ and δ^x only appears in the left-hand side. Since

$$[I - D\Phi_L(x^*)]\delta^x = [I - D\Phi(x^*)]\delta^x - [D\Phi_L(x^*) - D\Phi(x^*)]\delta^x,$$

the stability of the discretized problem is determined by the invertibility of the original infinite-dimensional problem—guaranteed by Assumption 5.4—provided that the linearization of the discretization in x^* approximates the linearization of the infinite-dimensional problem arbitrarily well. Indeed, this approximation holds for large L , as the following lemma states.

LEMMA 6.1 (consistency of derivative of Φ_L). *Let $x^* = (v^*, \alpha^*, \mu^*)$ be a fixed point of Φ , and let $G : C_\pi^0 \rightarrow \mathbb{R}^{n_y}$ be mildly differentiable once. Then there exists a monotone increasing continuous function $\omega^* : [0, \infty) \rightarrow [0, \infty)$ with $\omega^*(0) = 0$, such that*

$$(6.5) \quad \|D\Phi_L(x^*) - D\Phi(x^*)\|_{C_e^{0,1} \leftarrow C_e^{0,1}} = \|\mathcal{L}[I - \mathcal{P}_L]Dg(x^*)\|_{C_e^{0,1} \leftarrow C_e^{0,1}} \leq \omega^*(1/L).$$

Proof. The closed set $B_1^{0,1}(0)$ of functions with Lipschitz norm less than or equal to unity is compact in C_e^0 . By mild differentiability of g , the map $Dg(x^*)$ is in $\text{Lin}(C_e^0; C_e^0)$ and hence is a continuous (bounded) linear map from C_e^0 into itself. Consequently, the set $\mathcal{S} := \{Dg(x^*)y : \|y\|_{0,1} \leq 1\}$ is compact in C_e^0 and hence uniformly equicontinuous. This means that there exists a uniform modulus of continuity for this set $\mathcal{S}$, a continuous monotone increasing function $\omega_g : [0, \infty) \rightarrow [0, \infty)$ with $\omega_g(0) = 0$, such that

$$|[Dg(x^*)y](t+h) - [Dg(x^*)y](t)| \leq \omega_g(h) \quad \text{for all } t \in \mathbb{R} \text{ and } y \text{ with } \|y\|_{0,1} \leq 1.$$

Furthermore, the interpolation approximation $\mathcal{P}_L$ satisfies for any continuous function z (see [40])

$$(6.6) \quad \|z - \mathcal{P}_L z\|_\infty \leq 6(1 + \Lambda_m) \omega_z(C_{\text{msh}}/(2mL)),$$

where Λ_m is the Lebesgue constant for interpolation at the points $(t_{c,j})_{j=1}^m$ on the interval $[0, 1]$ chosen in the discretization (2.2), and $\delta \mapsto \omega_z(\delta)$ is the modulus of continuity for z . Thus, the equicontinuity with modulus of continuity ω_g on $\mathcal{S}$ implies that

$$(6.7) \quad \|[I - \mathcal{P}_L]Dg(x^*)y\|_\infty \leq 6(1 + \Lambda_m) \omega_g(C_{\text{msh}}/(2mL)) \quad \text{if } \|y\|_{0,1} \leq 1.$$

Defining $\omega^*(s) = 6(1 + \Lambda_m) \|\mathcal{L}\|_{C_e^{0,1} \leftarrow L_e^\infty} \omega_g(C_{\text{msh}} s/(2m))$, (6.7) implies the claim of the lemma. $\square$

Remark 6.2 (sharper estimate if G is mildly differentiable twice). If G is mildly differentiable twice, then $Dg(x^*) \in \text{Lin}(C_e^1; C_e^1)$, such that the sharper estimate

$$(6.8) \quad \|D\Phi_L(x^*) - D\Phi(x^*)\|_{C_e^{0,1} \leftarrow C_e^{0,1}} \leq \frac{3(1 + \Lambda_m)}{m} \frac{\|\mathcal{L}\|_{C_e^{0,1} \leftarrow L_e^\infty} \|Dg(x^*)\|_{C_e^1 \leftarrow C_e^1}}{L}$$

holds (replacing ω_g with $\|Dg(x^*)\|_{C_e^1 \leftarrow C_e^1}$).

The stability of the discretized problem is a straightforward consequence of the previous lemma.

COROLLARY 6.3. *Under Assumptions 5.1, 5.2, 5.4 from section 5, the fixed-point problem $\Phi_L(x) = x$ is stable in x^* ; that is, there exists a bound $C_{\text{stab},L}$ for $\|[I - D\Phi_L(x^*)]^{-1}\|_{C_e^{0,1} \leftarrow C_e^{0,1}}$ for sufficiently large L , and $C_{\text{stab},L} \rightarrow C_{\text{stab},\infty}$ for $L \rightarrow \infty$.*

Proof. By Assumption 5.4, $I - D\Phi(x^*)$ has a bounded inverse. The Banach perturbation lemma (see, e.g., [39, Theorem 2.1.1]) and Lemma 6.1 then ensure that $I - D\Phi_L(x^*) = I - \mathcal{L}\mathcal{P}_L Dg(x^*)$ has a bounded inverse, too, for sufficiently large L . Let L_{diff} be such that $\omega^*(1/L_{\text{diff}}) \leq 1/(2C_{\text{stab},\infty})$. Then, for $L \geq L_{\text{diff}}$, the inverse has the norm

$$(6.9) \quad \begin{aligned} \|[I - D\Phi_L(x^*)]^{-1}\|_{C_e^{0,1} \leftarrow C_e^{0,1}} &\leq \frac{C_{\text{stab},\infty}}{1 - \omega^*(1/L)C_{\text{stab},\infty}} \\ &\leq (1 + 2C_{\text{stab},\infty}\omega^*(1/L))C_{\text{stab},\infty} =: C_{\text{stab},L}. \end{aligned} \quad \square$$

A rough upper bound for the norm of the inverse of $I - D\Phi_L(x^*)$ for $L \geq L_{\text{diff}}$ is thus

$$(6.10) \quad C_{\text{stab}} := C_{\text{stab},L_{\text{diff}}} = (1 + 2C_{\text{stab},\infty}\omega^*(1/L_{\text{diff}}))C_{\text{stab},\infty} \leq 2C_{\text{stab},\infty}.$$

Using Corollary 6.3, we can isolate δ^x in the identity (6.2):

$$(6.11) \quad \delta^x = [I - \Phi_L(x^*)]^{-1}[\epsilon_c(L) + \epsilon_{\text{nl}}(L, \delta^x)].$$

In order to prove that the right-hand side of (6.11) is a contraction—and thus defines a well-posed fixed-point problem—we need suitable bounds for the consistency error $\epsilon_c(L)$ and the nonlinear part $\epsilon_{\text{nl}}(L, \delta^x)$. For the consistency error, the smoothness of the solution x^* , established in Corollary 5.3, enables us to apply the convergence theory for interpolation of smooth functions in the following lemma.

LEMMA 6.4. *Let x^* be a fixed point of Φ and G be mildly differentiable to order $\ell_{\text{max}} \geq 1$. Then the consistency error $\epsilon_c(L) = \mathcal{L}(\mathcal{P}_L - I)g(x^*)$ in (6.3) satisfies*

$$\|\epsilon_c(L)\|_{0,1} = O(L^{-\min\{\ell_{\text{max}}, m\}}),$$

where $m - 1$ is the degree of the interpolation polynomial used in $\mathcal{P}_L$ in (3.9).

Proof. By Corollary 5.3, x^* is in $C_e^{\ell_{\max}+1}$ and thus $g(x^*)$ is in $C_e^{\ell_{\max}}$. Let $[t_i, t_{i+1}]$ be one of the subintervals in the mesh of the discretization, where $i \in \{0, \dots, L-1\}$. If $\ell_{\max} \geq m$, then

$$(6.12) \quad \|(\mathcal{P}_L - I)g(x^*)|_{[t_i, t_{i+1}]}\|_\infty \leq \frac{\|[g(x^*)]^{(m)}|_{[t_i, t_{i+1}]}\|_0}{m!} [t_{i+1} - t_i]^m$$

follows by the Cauchy interpolation remainder theorem (see, e.g., [31, section 6.1, Theorem 2]). If $\ell_{\max} \leq m$, by [4, Theorems 1.8–1.9],

$$\|(\mathcal{P}_L - I)g(x^*)|_{[t_i, t_{i+1}]}\|_\infty \leq c \left[\frac{t_{i+1} - t_i}{2m} \right]^{\ell_{\max}} \|g(x^*)^{(\ell_{\max})}\|_0.$$

Inserting the upper bound C_{msh}/L for the interval lengths, the statement of the lemma follows from the boundedness of the operator $\mathcal{L}: L_e^\infty \rightarrow C_e^{0,1}$. $\square$

Smallness of nonlinear term $\epsilon_{\text{nl}}(L, \delta^x)$. The nonlinear term $\epsilon_{\text{nl}}(L, \delta^x)$ in the identity (6.2) is zero if δ^x is zero. We now want to find a Lipschitz constant for ϵ_{nl} with respect to δ^x that is sufficiently small. The following lemma provides us with an estimate.

LEMMA 6.5. *Let x^* be a fixed point of Φ and G be mildly differentiable to order $\ell_{\max} \geq 1$. Then for all $\rho_{\text{nl}} > 0$ there exists $r_{\text{nl}} > 0$ such that all $\delta^{x,1}, \delta^{x,2} \in \mathcal{B}_{r_{\text{nl}}}^{0,1}(0)$ and $L > 0$ satisfy*

$$\begin{aligned} \|\epsilon_{\text{nl}}(L, \delta^{x,1}) - \epsilon_{\text{nl}}(L, \delta^{x,2})\|_{0,1} &\leq \rho_{\text{nl}} \|\delta^{x,1} - \delta^{x,2}\|_{0,1}, \text{ implying, in particular,} \\ \|\epsilon_{\text{nl}}(L, \delta^x)\|_{0,1} &\leq \rho_{\text{nl}} \|\delta^x\|_{0,1} \quad \text{for all } \delta^x \in \mathcal{B}_{r_{\text{nl}}}^{0,1}(0). \end{aligned}$$

Proof. Mild differentiability of g implies by Corollary 4.2 that we can find for every $\rho_{\text{nl}} > 0$ a radius $r_{\text{nl}} > 0$ such that

$$(6.13) \quad \begin{aligned} &\|g(x^* + \delta^{x,1}) - g(x^* + \delta^{x,2}) - Dg(x^*)[\delta^{x,1} - \delta^{x,2}]\|_0 \\ &\leq \frac{\rho_{\text{nl}}}{\|\mathcal{L}\|_{C_e^{0,1} \leftarrow L_e^\infty} \|\mathcal{P}_L\|_{L_e^\infty \leftarrow C_e^0}} \|\delta^{x,1} - \delta^{x,2}\|_{0,1} \end{aligned}$$

for all $\delta^{x,1}, \delta^{x,2} \in \mathcal{B}_{r_{\text{nl}}}^{0,1}(0)$. In the denominator on the right-hand side, the norm $\|\mathcal{P}_L\|_{L_e^\infty \leftarrow C_e^0}$ has a uniform upper bound for all L as the degree $m-1$ of the interpolation polynomial in $\mathcal{P}_L$ in (3.9) is fixed. Note the stronger $\|\cdot\|_{0,1}$ -norm on the right-hand side in (6.13) required by mild differentiability. Thus, by the definition of ϵ_{nl} in (6.4),

$$\begin{aligned} &\|\epsilon_{\text{nl}}(L, \delta^{x,1}) - \epsilon_{\text{nl}}(L, \delta^{x,2})\|_{0,1} \\ &\leq \|\mathcal{L}\|_{C_e^{0,1} \leftarrow L_e^\infty} \|\mathcal{P}_L\|_{L_e^\infty \leftarrow C_e^0} \|g(x^* + \delta^{x,1}) - g(x^* + \delta^{x,2}) - Dg(x^*)[\delta^{x,1} - \delta^{x,2}]\|_0 \\ &\leq \rho_{\text{nl}} \|\delta^{x,1} - \delta^{x,2}\|_{0,1}. \end{aligned} \quad \square$$

Convergence result. We can now combine Corollary 6.3 and Lemmas 6.4 and 6.5 into a convergence theorem.

THEOREM 6.6 (high-order convergence of collocation). *Let G be mildly differentiable to order $\ell_{\max} \geq 1$, and let $x^* = (v^*, \alpha^*, \mu^*)$ be a fixed point of Φ , as defined in (3.5), with a bounded inverse of $I - D\Phi(x^*)$, as listed in Assumptions 5.1, 5.2, 5.4 in section 5.*

Then there exists a radius $r > 0$ such that the discretized fixed-point problem $x = \Phi_L(x)$ with L polynomial pieces of degree m has a unique solution x^L in the ball $B_r^{0,1}(x^*)$ for all sufficiently large L . The error $\delta^{x,L} = x^L - x^*$ satisfies

$$\|\delta^{x,L}\|_{0,1} = O(L^{-\min\{\ell_{\max}, m\}}).$$

Proof. After Lemmas 6.1, 6.4, and 6.5 have used mild differentiability to establish estimates for the ingredients of the splitting (6.2)–(6.4), the arguments in this proof are identical to those one would use for smooth right-hand sides, explained by [38] for a general situation. We include them here because they show the criteria for how large to choose the discretization level L .

We first claim that the map $h : C_e^{0,1} \rightarrow C_e^{0,1}$ given by the right-hand side of (6.11), i.e.,

$$h(\delta) = [I - D\Phi_L(x^*)]^{-1}[\epsilon_c(L) + \epsilon_{\text{nl}}(L, \delta)],$$

maps back into $B_r^{0,1}(0)$ and is a contraction for a sufficiently small r (which we need to find) and all sufficiently large L . By the identity (6.2), fixed points of h are fixed points of Φ_L .

Let $\kappa \in (0, 1)$ be arbitrary, and let us recall from the proof of stability in Lemma 6.1 the upper bound $C_{\text{stab}} > 0$ of $\|[I - D\Phi_L(x^*)]^{-1}\|_{C_e^{0,1} \leftarrow C_e^{0,1}}$ given in (6.10), which holds for all $L \geq L_{\text{diff}}$. We choose a radius $r = r_{\text{nl}}$ such that the factor ρ_{nl} in Lemma 6.5 satisfies $\rho_{\text{nl}} < \kappa/C_{\text{stab}}$. Then we choose the lower bound L_{min} for L ,

$$L_{\text{min}} = \max\{L_{\text{diff}}, L_c\}, \quad \text{where } L_c \text{ is s. t. } \|\epsilon_c(L)\|_{0,1} \leq \frac{1 - \kappa}{C_{\text{stab}}} r_{\text{nl}} \text{ for all } L \geq L_c.$$

For this choice of radius r and $L \geq L_{\text{min}}$, we have for all $\delta^x \in B_r^{0,1}(0)$

$$\begin{aligned} \|h(\delta^x)\|_{0,1} &\leq \|[I - D\Phi_L(x^*)]^{-1}\|_{C_e^{0,1} \leftarrow C_e^{0,1}} [\|\epsilon_c(L)\|_{0,1} + \rho_{\text{nl}}\|\delta^x\|_{0,1}] \\ (6.14) \quad &\leq C_{\text{stab}}\|\epsilon_c(L)\|_{0,1} + \kappa\|\delta^x\|_{0,1} \\ &\leq (1 - \kappa)r_{\text{nl}} + \kappa\|\delta^x\|_{0,1} \leq r_{\text{nl}} = r, \end{aligned}$$

such that h maps $B_r^{0,1}(0)$ back into itself. For the Lipschitz constant of h , we have by Lemma 6.5

$$\|\epsilon_{\text{nl}}(L, \delta^{x,1}) - \epsilon_{\text{nl}}(L, \delta^{x,2})\|_{0,1} \leq \rho_{\text{nl}}\|\delta^{x,1} - \delta^{x,2}\|_{0,1} \leq \frac{\kappa}{C_{\text{stab}}}\|\delta^{x,1} - \delta^{x,2}\|_{0,1}$$

for $\delta^{x,1}, \delta^{x,2} \in B_r^{0,1}$ such that by the definition of C_{stab} and Corollary 6.3 and Lemma 6.4 we have

$$\begin{aligned} \|h(\delta^{x,1}) - h(\delta^{x,2})\|_{0,1} &\leq \|[I - D\Phi_L(x^*)]^{-1}\|_{C_e^{0,1} \leftarrow C_e^{0,1}} \frac{\kappa}{C_{\text{stab}}}\|\delta^{x,1} - \delta^{x,2}\|_{0,1} \\ &\leq \kappa\|\delta^{x,1} - \delta^{x,2}\|_{0,1}. \end{aligned}$$

This implies that h has a unique fixed point in $B_r^{0,1}(0)$. Hence, by the identity (6.2), Φ_L has a unique fixed point in $B_r^{0,1}(x^*)$. Finally, inequality (6.14) implies for the fixed point $\delta^{x,L}$ of h that

$$\|\delta^{x,L}\|_{0,1} \leq \frac{C_{\text{stab}}}{1 - \kappa}\|\epsilon_c(L)\|_{0,1} = O(L^{-\min\{\ell_{\max}, m\}}). \quad \square$$

Precise bound of discretization error. Revisiting the definition (6.10) for the bound C_{stab} in Corollary 6.3, we observe that we can replace C_{stab} by the smaller $C_{\text{stab},L}$, which can be chosen as close to the infinite-dimensional stability constant $C_{\text{stab},\infty}$ as one wishes if one increases the discretization level L further. Similarly, we may choose the constant κ as close to 0 as desired, where again a smaller κ requires a larger lower bound on L . Furthermore, assuming that the order of mild differentiability $\ell_{\max}$ exceeds the order m of the discretization, we may insert the concrete estimate (6.12) for $\epsilon_c(L)$. We also observe that the m th time derivatives $[g(x^*)]^{(m)}$ equal the derivatives of order $m+1$ of v^* when we denote the first component of the solution x^* as v^* ($x^* = (v^*, \alpha^*, \mu^*)$). With these concrete estimates, we obtain that for every $\epsilon > 0$ there exists a lower bound $L_{\text{low}}(\epsilon)$ such that the error of the discretized fixed-point problem $\delta^{x,L}$ satisfies

$$(6.15) \quad \|\delta^{x,L}\|_{0,1} \leq (1 + \epsilon) C_{\text{stab},\infty} \frac{\|\mathcal{L}\|_{C_e^{0,1} \leftarrow L_e^\infty} \|(v^*)^{(m+1)}\|_0}{m!} \left[\frac{C_{\text{msh}}}{L} \right]^m$$

for all $L \geq L_{\text{low}}(\epsilon)$. This estimate is identical to the result one would obtain for discretizations of smooth nonlinear infinite-dimensional differential equations.

7. Implementation and numerical test. The numerical tool DDE-BIFTOOL permits implementation of systems of FDEs with finitely many discrete delays, which have the form

$$(7.1) \quad \begin{aligned} M\dot{y}(t) &= f(y(t - \tau_0), \dots, y(t - \tau_{n_d}), p), & \text{where } \tau_0 = 0, \text{ and} \\ \tau_j &= \tau_{\text{fun},j}(y(t - \tau_0), \dots, y(t - \tau_{j-1}), p) & \text{for } j = 1, \dots, n_d, \end{aligned}$$

and $M \in \mathbb{R}^{n_y \times n_y}$, $y(t) \in \mathbb{R}^{n_y}$, and $f: \mathbb{R}^{n_y \times (n_d+1)} \times \mathbb{R}^{n_p} \rightarrow \mathbb{R}^{n_y}$, $\tau_{\text{fun},j}: \mathbb{R}^{n_y \times j} \times \mathbb{R}^{n_p} \rightarrow \mathbb{R}$ are smooth functions of their arguments. The matrix M may be singular to permit the implicit definition of delays or the formulation of neutral FDEs (not analyzed in this paper). For this class of FDEs, the abstract right-hand side G_{FDE} in (1.1) has the form

$$(7.2) \quad \begin{aligned} G(y, p) &= f(y(-\tau_0), \dots, y(-\tau_{n_d}), p), & \text{where } \tau_0 = 0, \text{ and} \\ \tau_j &= \tau_{\text{fun},j}(y(-\tau_0), \dots, y(-\tau_{j-1}), p) & \text{for } j = 1, \dots, n_d. \end{aligned}$$

The functional G_{FDE} is mildly differentiable $\ell_{\max}$ times if the coefficient functions f and $\tau_{\text{fun},j}$ are $\ell_{\max}$ times continuously differentiable with respect to their arguments.

We perform our tests on the BVP (2.4)–(2.5) for $y_0 = 0.75$. As a starting guess for the Newton iterations, we choose the solution computed with DDE-BIFTOOL, unadapted mesh with $L = 200$ and $m = 7$. We recompute the solution using different values of L and m and approximate $\|y'/T + y(\cdot - p/T - y/T)\|_0$ by considering the maximum of the residuals on a uniform grid of 10001 points, as shown in the left plot of Figure 1. The picture on the right shows the (rescaled) solution obtained using $L = 10$ and $m = 5$, having actual period $T \approx 7.00$.

8. Conclusions. In the numerical bifurcation analysis of FDEs, there is often a strong interest in analyzing the long-term dynamics. Such analysis includes the detection and computation of equilibria and periodic orbits regardless of their dynamical stability in a continuation framework with respect to model parameters.

The present paper provided a complete and rigorous error analysis of the piecewise orthogonal collocation for computing periodic solutions of FDEs which may include state-dependent delays. Indeed, although the method has widely been used for two

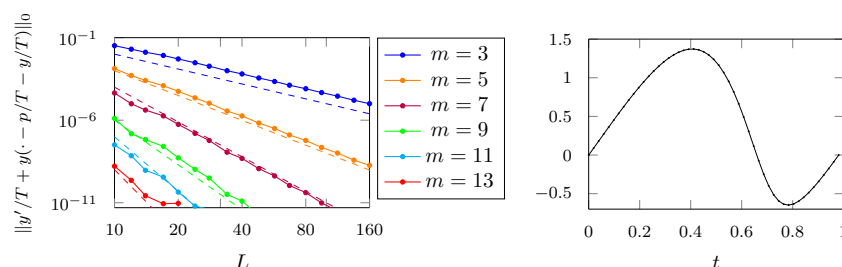


FIG. 1. Left: maximum among the errors computed at 10,001 equidistant points in $[0, T]$ for different values of m and L , compared to straight dashed lines having slope m . Right: periodic solution computed with $L = 10$ and $m = 5$, rescaled so as to have period 1.

decades [20, 45] and incorporated into software such as DDE-BIFTOOL and KNUT, the convergence of the corresponding finite-element method had only been supported by (many) numerical experiments but never proved theoretically for a general FDE.

A convergence analysis was recently performed for FDEs with constant delays [3], based on the general approach for BVPs established in [38]. The latter assumes regularity properties on the part of the right-hand sides that cannot be satisfied when state-dependent delays are present. However, using the concepts of weaker forms of differentiability developed for state-dependent delays [12, 27] allowed us to arrive at convergence estimates as sharp as those previously obtained for constant delays by only assuming to satisfy the regularity properties in the milder form.

In numerical bifurcation analysis, the next step after finding a periodic orbit numerically is consideration of its linear stability, which ensures local asymptotic stability with respect to compatible perturbations in the initial value [35]. For numerical computation of linear stability, one needs to approximate the monodromy operator and the Floquet multipliers. Although we are not aware of any theoretical study on the convergence of such approximation with respect to the FEM strategy, preliminary experimental evidence can be found in [8].

The analysis carried out does not immediately extend to the spectral approach, characterized by bounded L and increasing m , since it does assume in various points that the relevant interpolation operator is bounded. However, several of the apparent impediments to such an extension observed in [3] no longer hold once the BVP is formulated in periodic spaces of functions, as we have done in this paper. Therefore, the authors plan to reinvestigate the potential for spectral convergence for periodic BVPs of FDEs. Moreover, there is potential for extending the analysis to more general classes of delay equations, such as neutral FDEs, whose further restrictions on the regularity on the right-hand sides represent a substantial obstacle.

Another issue was left open in [3], even for the case of constant delays. Since the y -component of a numerical approximation is not differentiable for typical polynomial collocation schemes, it is unclear how the derivative of the right-hand side can be evaluated during Newton iterations, as it requires evaluation of $y'(t)$ at difficult-to-control times t . This discontinuous dependence on the solution implies that the standard convergence argument for Newton iterations needs to be revisited for FDEs.

Appendix A. Proof of Lemma 4.1. The extendability condition (condition 2) in Definition 2.1 is only formulated for the application of ℓ times the same deviation δ^u . The polarization identity (Proposition A.2 below) ensures that condition 2 also applies to ℓ different deviations.

LEMMA A.1 (multidirectional extendability). *If G is $\ell_{\max}$ times mildly differentiable according to Definition 2.1, then the map*

$$C^\ell \times (C^\ell)^\ell \ni (x, y^1, \dots, y^\ell) \mapsto D^\ell G(x) y^1 \dots y^\ell \in \mathbb{R}^{n_G}$$

can be extended to a continuous map in $C^\ell \times (C^{\ell-1})^\ell$ for all $\ell \in \{1, \dots, \ell_{\max}\}$.

PROPOSITION A.2 (polarization identity). *Let $\ell \geq 1$ be arbitrary. There exist 2^ℓ coefficients $a_i \in \mathbb{R}$ and $\ell 2^{\ell-1}$ coefficients $b_{i,j} \in \{-1, 1\}$ ($i \in \{1, \dots, 2^{\ell-1}\}$, $j \in \{1, \dots, \ell\}$) such that for all $c_1, \dots, c_\ell \in \mathbb{R}$ the following identity holds:*

$$\prod_{j=1}^{\ell} c_j = \sum_{i=1}^{2^{\ell-1}} a_i \left[\sum_{j=1}^{\ell} b_{i,j} c_j \right]^\ell.$$

Consequently, for any bounded ℓ -linear map M from $X \times \dots \times X$ to Y and arbitrary linear spaces X and Y , $Mx_1 \dots x_\ell = \sum_{i=1}^{2^{\ell-1}} a_i M[\sum_{j=1}^{\ell} b_{i,j} x_j]^\ell$ for all $x_1, \dots, x_\ell \in X$. In short, arbitrary combinations of arguments of the multilinear map can be expressed as linear combinations of the single-argument map $y \mapsto My^\ell$. For example, for $\ell = 2$, the coefficients are $a_1 = 1/4$, $a_2 = -1/4$, $b_{1,1} = b_{2,1} = b_{1,2} = 1$, $b_{2,2} = -1$: $Myz = \frac{1}{4}M[y+z]^2 - \frac{1}{4}M[y-z]^2$. Lemma A.1 follows by continuity.

Proof of Lemma 4.1. We show inductively over k that

$$C_\pi^{k+\ell} \times (C_\pi^{k+\ell-1})^\ell \ni (x, y^1, \dots, y^\ell) \mapsto D^\ell G_{\text{ts}}(x) y^1 \dots y^\ell = [t \mapsto D^\ell G(x_t) y_t^1 \dots y_t^\ell] \in C_\pi^k$$

is continuous if G is $k + \ell$ times mildly differentiable. The statement of Lemma 4.1 then follows from setting $k = \ell_{\max} - \ell$.

For $k = 0$, the fact that $D^\ell G_{\text{ts}}|_{C_\pi^\ell}$ is well defined at each time t follows from the assumption that $G|_{C_\pi^\ell}$ is ℓ times continuously differentiable by part 1 of Definition 2.1. The continuity with respect to the C_π^0 -norm in t follows from the fact that the time shift $\mathbb{R} \times C_\pi^j \ni (t, u) \mapsto u_t \in C_\pi^j$ is continuous for arbitrary $j \geq 0$.

For the inductive step, we first assume that for all integers $\ell \geq 0$ the function $t \mapsto D^\ell G(x_t) y_t^1 \dots y_t^\ell$ is in C_π^k for all $x \in C_\pi^{k+\ell}$ and all $y^1, \dots, y^\ell \in C_\pi^{k+\ell-1}$ if G is $k + \ell$ times mildly differentiable. We have to show that $D^\ell G(x_t) y_t^1 \dots y_t^\ell$ is in C_π^{k+1} for $x \in C_\pi^{k+\ell+1}$, $y^1, \dots, y^\ell \in C_\pi^{k+\ell}$ if G is $k + \ell + 1$ times mildly differentiable. The time difference quotient for $D^\ell G(x_t) y_t^1 \dots y_t^\ell$ at time t equals

$$(A.1) \quad \frac{1}{h} \left[D^\ell G(x_{t+h}) \prod_{i=1}^{\ell} y_t^i - D^\ell G(x_t) \prod_{i=1}^{\ell} y_t^i \right] + \sum_{j=1}^{\ell} D^\ell G(x_t) \left[\prod_{i \neq j}^{\ell} y_t^i \right] \frac{y_{t+h}^j - y_t^j}{h}.$$

The function $u \mapsto D^\ell G(u) \prod_{i=1}^{\ell} y_t^i$ is continuously differentiable with respect to u in $u = sx_{t+h} + (1-s)x_t$ for $s \in [0, 1]$ since $x \in C_\pi^{k+\ell+1}$, $y^1, \dots, y^k \in C_\pi^{k+\ell}$ and $k \geq 1$ by the assumption that G is $k + \ell + 1$ times mildly differentiable and $u \in C_\pi^{k+\ell+1}$ and $y^1, \dots, y^\ell \in C_\pi^{k+\ell}$ with $k + \ell \geq \ell + 1$. Thus, we may apply the mean-value theorem for the first term in (A.1), resulting in the expression

$$\int_0^1 D^{\ell+1} G(sx_{t+h} + (1-s)x_t) \frac{x_{t+h} - x_t}{h} \prod_{i=1}^{\ell} y_t^i ds.$$

By Lemma A.1, the derivative $D^{\ell+1} G(u) v_0 \dots v_\ell$ is continuous for arguments $u \in C_\pi^{\ell+1}$, $v_0, \dots, v_\ell \in C_\pi^\ell$. The argument $u = sx_{t+h} + (1-s)x_t$ is in $C_\pi^{k+\ell+1}$, $v_0 = [x_{t+h} - x_t]/h$ is in $C_\pi^{k+\ell+1}$, and y_t^i and the limit x_t' of $[x_{t+h} - x_t]/h$ are in $C_\pi^{k+\ell}$ such that we can take the limit for $h \rightarrow 0$ to obtain

$$\lim_{h \rightarrow 0} \frac{1}{h} \left[D^\ell G(x_{t+h}) \prod_{i=1}^{\ell} y_t^i - D^\ell G(x_t) \prod_{i=1}^{\ell} y_t^i \right] = D^{\ell+1} G(x_t) x'_t \prod_{i=1}^{\ell} y_t^i.$$

The right-hand side is a C_π^k function by the assumption of the inductive step.

In the second sum in (A.1), the difference quotient $[y_{t+h}^j - y_t^j]/h$ has the limit $(y_t^j)'$ in $C_\pi^{k+\ell-1}$ and the derivative $D^\ell G(x_t)$ can be extended continuously to multilinear arguments in $C_\pi^{k+\ell-1}$ since G is $k + \ell \geq \ell$ times mildly differentiable. Thus, the limit for $h \rightarrow 0$ also exists for the second sum in (A.1):

$$(A.2) \quad \lim_{h \rightarrow 0} D^\ell G(x_t) \prod_{i \neq j}^{\ell} y_t^i \frac{y_{t+h}^j - y_t^j}{h} = D^\ell G(x_t) \prod_{i \neq j}^{\ell} y_t^i (y_t^j)'.$$

The argument x in this limit is in $C_\pi^{k+\ell+1}$, the arguments y_t^i are in $C_\pi^{k+\ell}$, and $(y^j)'$ is in $C_\pi^{k+\ell-1}$. Thus, by the assumption in the inductive step the limits in (A.2) are also in C_π^k for $j = 1, \dots, \ell$. This implies that the limits of both terms in (A.1) and thus the limit of the time difference quotient (A.1) are in C_π^k . Hence, $t \mapsto D^\ell G(x_t) y_t^1 \dots y_t^\ell$ is in C_π^{k+1} . Finally, we check the continuity of the map $C_\pi^{k+\ell+1} \times (C_\pi^{k+\ell})^\ell \ni (x, y^1, \dots, y^\ell) \mapsto D^\ell G_{ts}(x) y^1 \dots y^\ell \in C_\pi^{k+1}$. By inductive assumption, the map is continuous as a map into C_π^k . Inspecting (A.1) and (A.2), its derivative in t is

$$D^{\ell+1} G(x_t) x'_t \prod_{i=1}^{\ell} y_t^i + \sum_{j=1}^{\ell} D^\ell G(x_t) \prod_{i \neq j}^{\ell} y_t^i (y_t^j)'.$$

All terms in this sum are continuously mapping $C_\pi^{k+\ell+1} \times (C_\pi^{k+\ell})^\ell \ni (x, y^1, \dots, y^\ell)$ into C_π^k by inductive assumption, such that $(x, y^1, \dots, y^\ell) \mapsto D^\ell G_{ts}(x) y^1 \dots y^\ell$ is continuous into C_π^{k+1} (end of proof for Lemma 4.1).

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